

let $f \in C^0(\mathbb{R} \times \omega)$ with compact support $\text{supp}(f)$. then:

$$\exists c \geq 0, \forall t \in [0, 1], \quad \int_{\omega} |f(t, x)|^2 dx \leq c$$

proof

1. spatial support: $k = \pi_2(\text{supp}(f)) \subset \omega$ is compact. for $x \notin k$, $f(t, x) = 0$.
2. integrability: for each t , $x \mapsto |f(t, x)|^2$ is continuous with compact support k , hence $f(t, \cdot) \in L^2(\omega)$.
3. continuity: by continuous dependence under compact support, $g(t) = \int_{\omega} |f(t, x)|^2 dx$ is continuous on $[0, 1]$.
4. bound: by extreme value theorem on $[0, 1]$, $g(t)$ is bounded by some $c \geq 0$.